Hospital ERP SaaS — Decisions, Open Questions & Assumptions

Oct 12, 2026 · @Bytespate Limited

Thirty decisions are CONFIRMED (C-001..C-030), 22 engineering proposals await approval (P-001..P-022), six items are DEFERRED, and 66 questions remain OPEN with safe interim handling. This register is the single authoritative place for decisions: other documents cite these IDs, and if any document conflicts with this register, the register wins and the conflict is logged as a contradiction.

Owner: product owner (business decisions) and technical architect (engineering proposals). Version 0.4.0; all v0.3 IDs are kept unchanged. Open questions and assumptions/contradictions have their own tabs.

Decisions needed next

The first release option and the pilot hospital profile unblock everything else; these five groups are listed in priority order.

1. **Release option (Q-001) and pilot hospital profile (Q-042).** Plan for RP-A Outpatient & Diagnostics unless the pilot needs inpatient care on day one (RP-B).
2. **Release gate RG-0 approvals:** ADR-001 (modular monolith), ADR-002 (Next.js usage), ADR-003 (tenancy/RLS), ADR-004 (auth), ADR-005 (contracts), ADR-006 (outbox/jobs, plus the Redis 8 vs Valkey licence choice), ADR-012 (CI gates), ADR-013 (Jest), P-021 (version pins).
3. **First-slice business answers (front desk + cashier):** Q-003 roles; Q-020 rounding; Q-022 overpayment; Q-024 one payment across two invoices; Q-034 MRN format; Q-035 required fields and scripts; Q-037 queue/capacity; Q-038 doctor fee rules; Q-039 who issues invoices; Q-046 printers; Q-050 digits, dates and print languages; Q-051 MFA and timeouts; Q-068 method-correction reporting day; and **Q-004 deposits**, or confirm deposits are excluded at first.
4. **Operations:** Q-043 RPO/RTO and support hours; Q-044 VPS provider and region; Q-055 GitHub plan; Q-056 SonarQube host.
5. **Capacity meaning:** Q-041 — does "100,000 users" mean registered, daily active or concurrent users?

If the person who answered the planning questions is not the product owner, the product owner should also ratify C-022..C-025.

1. Status vocabulary

Label	Meaning	Who may change it
CONFIRMED	Settled by the product owner or by the requester in a recorded planning conversation.	Product owner only, through the amendment procedure (§5).
SOURCE-DERIVED	Taken from one of the supplied hospital documents S1–S7. It describes one hospital's needs and becomes product behavior only after an adoption decision. v0.3 called this label `SOURCE`.	Domain lead proposes adopt / configure / defer / reject; product owner approves.
PROPOSED	An engineering or planning recommendation in this pack. It must not be implemented as if approved, except inside a time-boxed spike.	The technical architect proposes; the product owner or tech lead approves.
OPEN	An unanswered question. Agents must not choose an answer (Open questions tab).	Answered by the named responder.
DEFERRED	Deliberately parked. It must not be silently implemented or silently ruled out.	Product owner.

2. Confirmed decisions

2.1 Carried forward from v0.3 (wording preserved)

ID	Decision	Implication for design	Source
C-001	Whole-hospital ERP offered as SaaS to multiple customers.	Customer data and configuration are separated. One maintained product.	v0.3 conversation
C-002	Bangladesh is the initial market.	Capture local operational conventions. Government discussion is deferred (C-021).	v0.3

ID	Decision	Implication for design	Source
C-003	Stack direction: NestJS + Next.js + PostgreSQL + Turborepo.	The backend API owns business writes. Monorepo boundaries apply.	v0.3
C-004	AI agents drive software development.	Tasks need explicit specifications and evidence (Implementation Plan G).	v0.3
C-005	Main journey: registration → visit → service order → bill → payment → delivery → outcome → next visit.	Keep this order, but model the lifecycles separately.	v0.3
C-006	Appointment, walk-in and direct diagnostic visits are all supported.	A diagnostic visit does not need a doctor consultation.	v0.3
C-007	An appointment booking and the actual visit are distinct.	Check-in creates or links the visit.	v0.3
C-008	One combined outpatient service invoice per visit. The doctor's visit fee is issued on a separate invoice.	Independent invoices, balances and receipts.	v0.3
C-009	The cashier collects the doctor fee.	Access to the doctor worklist does not grant cashier authority.	v0.3
C-010	Cash, card, mobile payment and bank transfer are supported.	Each payment keeps its actual method.	v0.3
C-011	Payment comes before service. Both full payment and permitted partial/deposit payment are supported.	Service eligibility is configurable. Emergency exceptions are OPEN (Q-005).	v0.3
C-012	A patient may use the hospital pharmacy or an outside pharmacy.	Outside fulfilment never changes hospital stock.	v0.3
C-013	Admission can come from an OPD referral, an emergency referral, or a planned/direct route.	Keep the admission source and the linked visit.	v0.3

ID	Decision	Implication for design	Source
C-014	An admission may wait for a bed.	Admission status and bed occupancy are separate.	v0.3
C-015	Bed transfers are staff-controlled and keep history.	The software records movement. It does not decide clinical need.	v0.3
C-016	After discharge, the vacated bed must be cleaned before it is available again.	Completing cleaning is an authorized operation.	v0.3
C-017	IPD uses a running account, deposits/partial payments and final billing.	Final settlement policy is OPEN (Q-006).	v0.3
C-018	A next appointment may be called a "next visit".	The UI glossary distinguishes the booking from the attended visit.	v0.3
C-019	Lab processing includes review/release and tracking of rejection and recollection.	Exact reviewer authority is OPEN (Q-009).	v0.3
C-020	Existing decisions take precedence over the supplied documents.	Conflicts are flagged, and an explicit amendment is sought.	v0.3
C-021	Government-specific rules are deferred for this discussion.	Earlier legal notes are not turned into current requirements (DEF-001).	v0.3

Identity and workflow agreements carried forward from v0.3 (CONFIRMED):

- Patient search comes before new registration.
- A possible duplicate produces a warning and is never merged automatically.
- MRN generation belongs to the hospital/customer scope.
- NID is optional.
- An unknown emergency patient can have a temporary identity, reconciled later by an authorized user.
- Direct diagnostic ordering is configured per test.
- Staff assign emergency priority, and the software records who assigned it and when.

These appear as requirements in SRS modules C02 and C10.

2.2 New confirmed decisions (planning conversation of 2026-10-12)

The requester answered these during the documentation task. If the requester is not the product owner, the product owner should ratify C-022..C-025 at the next review; until then they are treated as CONFIRMED.

ID	Decision	Implication	Replaces / resolves
C-022	Source code is hosted on **GitHub**, and CI/CD runs on **GitHub Actions**.	Pipeline templates are for GitHub Actions; images go to GHCR. The GitHub plan is still OPEN (Q-055) and affects environment protection rules.	v0.3 "CI/CD, provider OPEN"
C-023	Code quality uses **self-hosted SonarQube Community Build** (free).	Community Build analyzes only the main branch, so pull requests cannot be gated by Sonar. PR-level quality uses ESLint with the SonarJS plugin plus test and coverage checks. The Sonar quality gate on `main` blocks promotion to release (ADR-012).	v0.3 "SonarQube concrete gate OPEN"
C-024	Initial customer model: **one hospital per customer organization (tenant)**. Multi-facility groups come later.	The data model keeps the tenant → facility layer, but each tenant starts with exactly one facility. MRN scope is the tenant. Group features are DEFERRED (DEF-002).	**Answers Q-002 for the initial product.** Group behavior stays deferred.

ID	Decision	Implication	Replaces / resolves
C-025	Staff interface is **bilingual Bangla and English, switchable per user**, with **bilingual printouts**.	Every staff screen, message and print template needs both languages. Names and addresses support Bangla script. Per-document print-language rules and numeral style are OPEN (Q-050).	v0.3 "Bangla/English scope OPEN"

2.3 Confirmed by the 2026-10-12 documentation brief

ID	Decision	Implication
C-026	Deployment is designed for **VPS hosting**. No provider, server size or production topology has been selected.	VPS Operations (J) describes options and a single-VPS start with an expansion path. Nothing names a provider as chosen.
C-027	AI agents help **develop** the software. The product has **no autonomous clinical AI** features.	No automated diagnosis, triage, dosing or result interpretation. Clinical decisions remain with authorized staff.
C-028	The "100,000 users" figure is an **unresolved capacity-planning question**, not a validated target or guarantee.	The Capacity Plan (M) models registered, daily-active and concurrent scenarios. No document may claim support for 100k users without measured evidence (Q-041).
C-029	Baseline integrity requirements are mandatory: tenant isolation; server-side permission enforcement; audit of sensitive actions; clinical-record integrity (released or signed records are corrected, never overwritten); financial correctness (exact amounts, no duplicate postings, reconciliation).	These apply to every module, even while government rules are deferred. Mechanisms such as RLS are PROPOSED; the obligations are CONFIRMED.
C-030	Old delivery-duration and team-size estimates are **not** commitments.	No calendar dates, durations, team sizes or productivity multipliers appear in the pack.

3. Proposed engineering decisions

All 22 entries are **PROPOSED** until the named approver accepts them. P-001..P-006 keep their v0.3 meaning; later entries refine or add to them. Each consequential proposal has an ADR (Architecture doc, ADRs tab).

ID	Proposal	ADR	Approval needed / evidence
P-001	Modular monolith: one NestJS API process plus a separate worker process sharing domain packages.	ADR-001	Tech lead approval. The scaffold spike shows module boundaries enforced by lint.
P-002	Tenant organization with facilities beneath it.	ADR-003	**Partly settled by C-024.** One facility per tenant initially; the layer stays in the model.
P-003	Shared PostgreSQL schema with `tenant_id` on every tenant-owned row, composite tenant foreign keys, and row-level security as a second barrier. Tenant context is set transaction-locally.	ADR-003	The spike proves RLS with Prisma 7 + `pg` pool + PgBouncer (if used), using the real application role.
P-004	One product codebase with versioned, effective-dated customer configuration (policy versions). No customer-specific branches.	ADR-001	Product owner confirms the configurable-policy catalogue (SRS Overview §6).
P-005	Deliver in complete patient-journey slices.	None	Product owner chooses a release boundary (Q-001; BRD §8).
P-006	Supporting libraries: TanStack Query, React Hook Form, TanStack Table, Tailwind CSS, BullMQ, Redis-compatible store, S3-compatible storage, OpenAPI, OpenTelemetry.	Stack assessment	Assessment complete. Each library is approved by the tech lead after the scaffold spike.

ID	Proposal	ADR	Approval needed / evidence
P-007	Use Next.js (confirmed, C-003) as a staff workspace: client-rendered screens with TanStack Query, a server-rendered shell, **no Server Actions or Route Handlers that write business data**, and `output: 'standalone'`.	ADR-002	Tech lead approval.
P-008	First-party authentication owned by NestJS: opaque server-side sessions in httpOnly cookies; argon2id password hashing; TOTP second factor for privileged roles; account lockout with a visible reason.	ADR-004	Product owner confirms MFA scope and timeouts (Q-051).
P-009	Zod schemas in `packages/contracts` are the canonical API and event contract. OpenAPI is generated from them, and Prisma types never cross the API boundary.	ADR-005	The spike selects the OpenAPI generator compatible with NestJS 12 + Zod 4.
P-010	Transactional outbox, inbox deduplication and idempotency records in PostgreSQL. BullMQ on a Redis-protocol server delivers and schedules jobs; PostgreSQL stays the source of truth.	ADR-006	Tech lead approval. Choose Redis 8 or Valkey after the compatibility spike.
P-011	Money is stored as PostgreSQL `numeric(18,2)` with an explicit currency (BDT default). Domain arithmetic uses integer minor units, and rounding is explicit per policy.	ADR-007	The finance reviewer answers the rounding questions (Q-020).
P-012	Bed occupancy is stored as half-open time intervals (`tstzrange`) with PostgreSQL exclusion constraints.	ADR-008	Tech lead approval. The extra-bed policy (Q-007) may add rules.

ID	Proposal	ADR	Approval needed / evidence
P-013	Posted financial facts, stock movements, released results and signed clinical documents are append-only. Changes happen through correction, reversal or amendment records.	ADR-009	Finance and clinical reviewers agree on the correction flows.
P-014	Printing: browser print views (HTML/CSS) for routine desk output; server-generated PDF from the same template and record version, rendered by headless Chromium in an isolated container, for archived, reprinted or delivered documents.	ADR-010	UAT on the customer's real printers (Q-046).
P-015	Production starts on a single VPS running Docker Compose: Caddy reverse proxy, web, API, worker, PostgreSQL, Redis-compatible store. Object storage and backups are kept off the VPS. A documented expansion path follows.	ADR-011	Product owner accepts the single-host availability risk for the pilot, or chooses an expanded topology (Q-043).
P-016	CI gates: PRs are blocked by lint (including SonarJS rules), typecheck, tests, coverage thresholds, migration checks and security scans; SonarQube Community analysis of `main` gates promotion to release.	ADR-012	Tech lead approval. Confirm the GitHub plan (Q-055).
P-017	Keep Jest (supplied foundation) for unit and API tests on Node ⩾ 24.9 with ESM. Adopt Vitest only by an explicit decision if the Jest + ESM spike fails.	ADR-013	Tech lead; resolves Q-054.

ID	Proposal	ADR	Approval needed / evidence
P-018	Object storage through the S3 API. Do not adopt MinIO Community Edition for new production use (binary and image distribution stopped; repository archived). Use provider object storage or a maintained self-hosted alternative (SeaweedFS, Garage).	ADR-014	DevOps lead. Choose after the provider is selected (Q-044).
P-019	Worklists refresh by polling with short intervals and conditional requests. Server-Sent Events are a later optimization when measurements justify them.	ADR-015	Tech lead. Revisit with load-test evidence (Capacity Plan M).
P-020	Payment allocation is recorded at invoice-line level so per-service eligibility can be computed. The *policy* for choosing lines stays OPEN (Q-004, Q-024).	ADR-016	The finance reviewer answers Q-004.
P-021	Version pins for the scaffold: Node 24 LTS now (re-evaluate Node 26 after its LTS promotion); TypeScript 6.0.x; NestJS 12.1.x; Next.js 16.4.x plus the 2026-10-14 security release; Prisma 7.10.x (CLI and client together); PostgreSQL 18.x current minor; pnpm (exact version recorded in `packageManager`); Turborepo 2.11.x.	Stack & versions	Re-verify on the scaffold date and record the lockfile commit.
P-022	Bilingual implementation: `next-intl` message catalogs (`bn-BD`, `en`); self-hosted Noto Sans Bengali / Noto Sans fonts; print templates carry a language per output; stored person names keep the script as entered.	ADR-017	Native-speaker review of terminology (Q-050).

4. Deferred and superseded items

Six items are deliberately parked; none may be silently built or silently ruled out. **DEF-001 is not a completed compliance assessment.**

ID	Item	Why deferred	Revisit trigger
DEF-001	Government-specific requirements: statutory rules, mandatory reports, national health-record exchange, legally required hosting/retention.	Owner request (C-021).	Owner opens a separately scoped review. Its absence is **not** a completed compliance assessment.
DEF-002	Multi-facility hospital groups: shared patients across facilities, group-level reporting, cross-facility transfers, facility-specific MRN series.	C-024.	First customer with more than one facility.
DEF-003	Offline operation and synchronization.	Not requested; a large design commitment.	Pilot downtime discussion (Q-043) shows paper fallback is insufficient.
DEF-004	Full PACS/archive implementation.	A radiology module does not imply a PACS.	A customer requires DICOM storage or viewing (Q-028).
DEF-005	Dedicated per-customer deployments and database-per-tenant placement.	Shared tenancy is the proposed standard (P-003).	A contractual or capacity need (Capacity Plan M §8).
DEF-006	SaaS subscription billing, plans and commercial lifecycle automation.	Commercial decisions are OPEN (Q-052).	Before onboarding a second paying customer.

Superseded or partly answered v0.3 items:

Earlier item	Now	Note
v0.3 Q-002 (one hospital or group?)	Answered for the initial product by **C-024**.	Group behavior → DEF-002.

Earlier item	Now	Note
v0.3 "Bangla/English scope OPEN"	**C-025**.	Detail questions remain in Q-050.
v0.3 "CI/CD provider OPEN"	**C-022**.	None
v0.3 "SonarQube concrete gate OPEN"	**C-023** + P-016.	None

5. How decisions flow, and how they change

An open question, an adopted source requirement or an approved proposal becomes a confirmed decision, which then drives requirements, tests, tasks and release gates.

```
flowchart LR
  Q[Open question] -->|answered by named responder| C[Confirmed decision]
  S[Source requirement SRC-*] -->|adoption review| C
  P[Proposal P-* / ADR] -->|approved| C
  C --> R[Requirement FR-*/NFR-* in the SRS]
  R --> T[Tests AT-*/UAT-*]
  R --> G[Implementation tasks]
  T --> N[Traceability and release gates]
```

Amendment procedure:

1. Keep stable IDs. Never reuse or renumber an ID.
2. A change to a CONFIRMED item creates a new row (for example `C-008a`) or a new ID with a "Replaces" note. Record the date, approver, reason, replaced text, affected workflows, requirements, data, contracts and tests, and migration implications.
3. Mark the old row "Superseded by …". Do not edit its wording to look as if it was always agreed.
4. Update the traceability register and any affected ready tasks in the same change.
5. ADRs keep their alternatives and consequences after approval. An ADR is superseded by a later ADR, never deleted.

Decision record template:

```
ID:            C-0NN | P-0NN | DEF-0NN
Title:
Status:        CONFIRMED | PROPOSED | DEFERRED | SUPERSEDED
Date / approver:
Context (link question/source):
Decision:
Consequences (data, contracts, UI, tests, operations):
Replaces:
Linked requirements / ADR / tests:
```